

学号： 2106410101



沈阳建筑大学

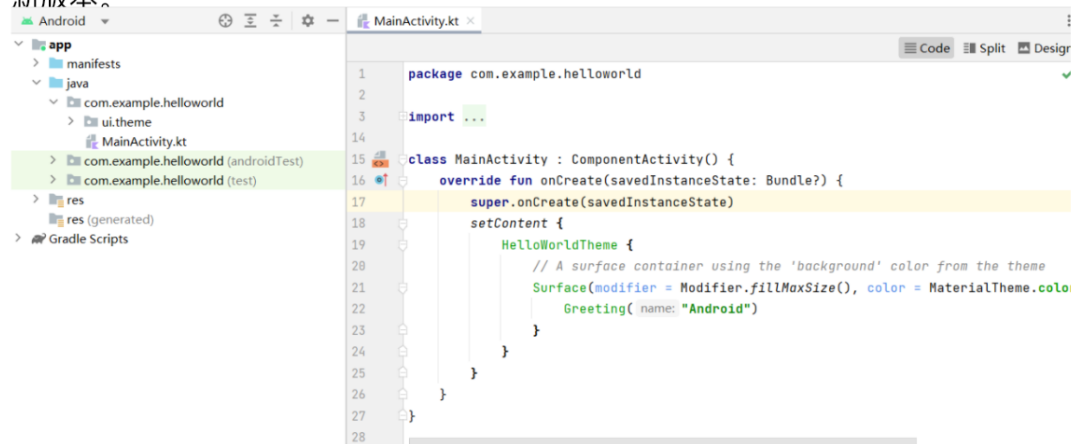
智能手机应用程序开发 课后作业报告

2022-2023年秋季学期

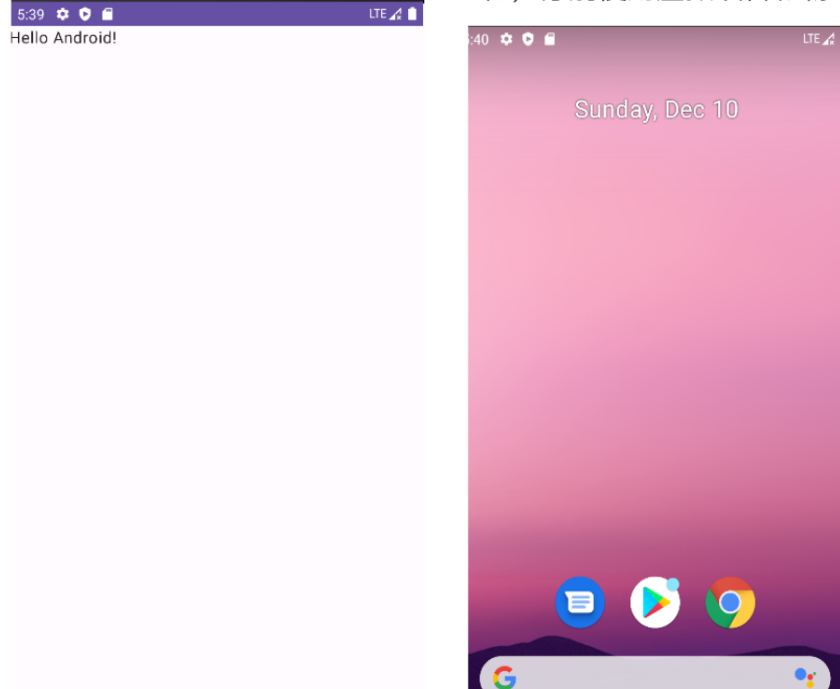
学生姓名： 马胜涛 专业班级： 计算机2101
指导教师： 郑新录 成 绩：

第一章作业：

1. 下载最新版本的 Android Studio, 安装后如果需要更新 Gradle, 则将其更新到最新版本。



2. 创建一个任意类型的手机 App 工程, 分别使用虚拟设备和物理设备运行。



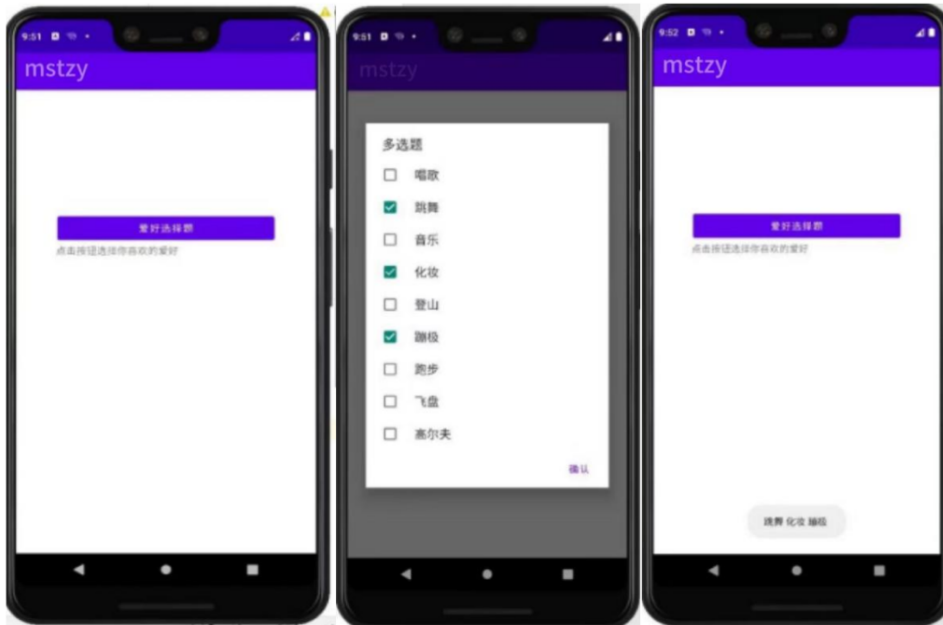
第二章作业：

1.实现多选题的界面效果。

部分关键代码：

```
String strList[] = new String[] {"唱歌","跳舞","音乐","化妆","登山","蹦极","跑步","飞盘","高尔夫"};
boolean blList[] = new
boolean[]{false,false,false,false,false,false,false,false,false};
private void cancelBool(){
    for(short
index=0;index<blList.length;++index){blList[index] = false;
}
} public void on2click(View view) {
    Button button =
    findViewById(R.id.button);
    if(button != null){
        AlertDialog.Builder aBuilder = new
        AlertDialog.Builder(MainActivity.this); aBuilder.setTitle("多选题");
        aBuilder.setMultiChoiceItems(strList, blList, new
        DialogInterface.OnMultiChoiceClickListener() { public void
        onClick(DialogInterface dialogInterface, int i, boolean b) {blList[i] =
        b; }});
        aBuilder.setPositiveButton("确认", new DialogInterface.OnClickListener()
        {
        public void onClick(DialogInterface dialogInterface, int i) {
            for(short
            index=0;index<blList.length;++index
            ){ if(blList[index]){ txt +=
            (strList[index]+" ");}
            }
            Toast.makeText(MainActivity.this, txt,
            Toast.LENGTH_SHORT).show();
            cancelBool();} });
        aBuilder.setOnCancelListener(new
        DialogInterface.OnCancelListener() { public void
        onCancel(DialogInterface dialogInterface) {
        cancelBool();}});
        aBuilder.show(); }}
```

运行结果：



2.实现选择出生日期的界面效果。部分关键代码:

```
int year,month,day;
```

```
DatePicker
```

```
datePicker;
```

```
@Override
```

```
protected void onCreate(Bundle savedInstanceState) {
```

```
    super.onCreate(savedInstanceState);
```

```
    setContentView(R.layout.activity_main);
```

```
    datePicker=findViewById(R.id.datePicker); Calendar
```

```
    calendar=Calendar.getInstance();
```

```
    year=calendar.get(Calendar.YEAR);
```

```
    month=calendar.get(Calendar.MONTH);
```

```
    day=calendar.get(Calendar.DAY_OF_MONTH);
```

```
    datePicker.init(year, month, day, new
```

```
    DatePicker.OnDateChangedListener() { public void
```

```
    onDateChanged(DatePicker datePicker, int year1, int month1, int day1)
```

```
    { MainActivity.this.year=year1;
```

```
        MainActivity.this.month=month1;
```

```
        MainActivity.this.day=day1;
```

```
        show(year,month,day); } };
```

```
private void show(int i, int i1, int i2) {
```

```
    String str="你的生日是: "+i+"年"+(1+i1)+"月"+i2+"日";
```

```
    Toast.makeText(MainActivity.this,str,Toast.LENGTH_SHORT).show();}
```

运行结果:



第三章作业:

- 1、实现注册界面的布局效果, 至少包括用户名、密码、忘记密码、登录等控件。

部分代码展示:

登录界面代码:

```
public static final int RESULT_CODE_REGISTER
= 0; private Button btnRegister;
private EditText
etAccount,etPass,etPassConfirm; private
CheckBox cbAgree;
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_register);
    getSupportActionBar().setTitle("注册");
    etAccount = findViewById(R.id.et_account);
    etPass = findViewById(R.id.et_password);
    etPassConfirm = findViewById(R.id.et_password_confirm);
    cbAgree = findViewById(R.id.cb_agree);
    btnRegister = findViewById(R.id.btn_register);
    btnRegister.setOnClickListener(this);}

public void onClick(View v){String name = etAccount.getText().toString();
    String pass = etPass.getText().toString();
    String passConfirm = etPassConfirm.getText().toString();
    if (TextUtils.isEmpty(name)) {
        Toast.makeText(RegisterActivity.this, "
        用户名不能为空", Toast.LENGTH_LONG).show();return; }
    if (TextUtils.isEmpty(pass)) {Toast.makeText(RegisterActivity.this, "密码不
    能为空", Toast.LENGTH_LONG).show();return;}
    if (!TextUtils.equals(pass,passConfirm))
    {Toast.makeText(RegisterActivity.this, "密码不一致",
    Toast.LENGTH_LONG).show(); return; }
    if (!cbAgree.isChecked()) {Toast.makeText(RegisterActivity.this, "请同意用
    户协议 ", Toast.LENGTH_LONG).show();return; }
    SharedPreferences spf = getSharedPreferences("spfRecord",
    MODE_PRIVATE);
    SharedPreferences.Editor edit = spf.edit();
    edit.putString("account", name);
    edit.putString("password", pass);
    edit.apply();
    Intent intent = new Intent();
    Bundle bundle = new Bundle();
    bundle.putString("account",name);
    bundle.putString("password",pass);
    intent.putExtras(bundle);
    setResult(RESULT_CODE_REGISTER,intent);
    Toast.makeText(RegisterActivity.this, "注册成功! ",
    Toast.LENGTH_LONG).show(); this.finish(); }}

运行结果:
```



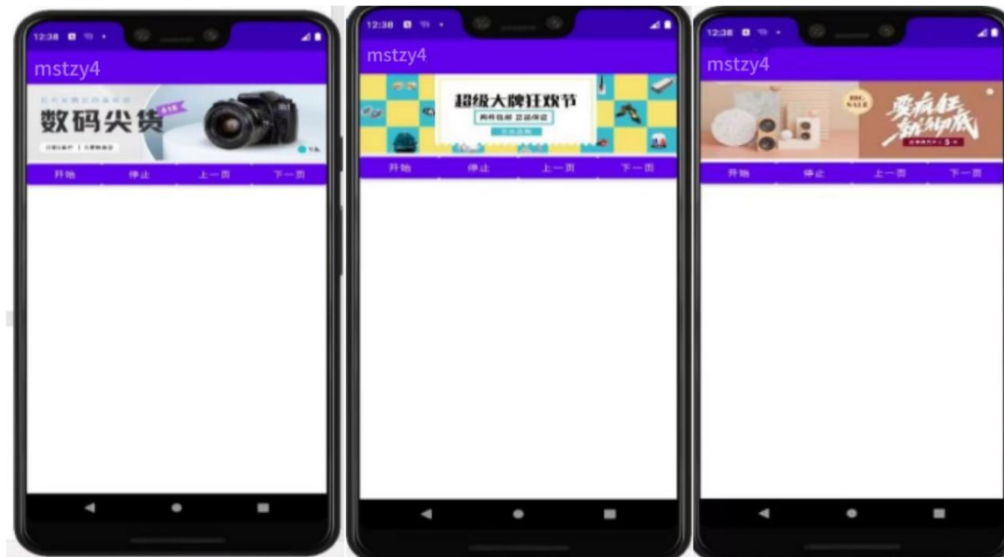
第四章作业：

1. 实现京东首页的轮播广告效果，至少包括 3 个产品广告的图片。

部分代码展示：

```
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    View view3 = View.inflate(MainActivity.this,
        R.layout.view_flipper_page3, null); ImageView imageView4
    = new ImageView(this);
    imageView4.setAdjustViewBounds(true);
    imageView4.setImageResource(R.mipmap.img_95
        46); mViewFlipper =
    findViewById(R.id.view_flipper);
    mViewFlipper.addView(imageView4);
    mViewFlipper.addView(view3,2);
    mViewFlipper.startFlipping();
    mViewFlipper.setOnClickListener(new View.OnClickListener() {
        public void onClick(View v) {Toast.makeText(MainActivity.this, "当
    前显示的子视图为"+mViewFlipper.getDisplayedChild(),
    Toast.LENGTH_SHORT).show();} });
    Button startBtn =
    findViewById(R.id.button_start);
    startBtn.setOnClickListener(new
    View.OnClickListener() { public void
    onClick(View v)
    {mViewFlipper.startFlipping();}}); Button
    stopBtn = findViewById(R.id.button_stop);
    stopBtn.setOnClickListener(new
    View.OnClickListener() { public void
    onClick(View v)
    {mViewFlipper.stopFlipping();}}); Button
    previousBtn =
    findViewById(R.id.button_previous);
    previousBtn.setOnClickListener(new
    View.OnClickListener() { public void
    onClick(View v)
    {mViewFlipper.showPrevious();}}); Button
    nextBtn = findViewById(R.id.button_next);
    nextBtn.setOnClickListener(new
    View.OnClickListener() { public void
    onClick(View v) {mViewFlipper.showNext();}});}
```

运行结果：



第五章作业：

- 1、实现登录后显示用户名的效果。输入用户名、密码后登录，启动一个新的Activity，显示用户名
部分关键代码：

```
private static final String TAG = "tag";
```

```

public static final int REQUEST_CODE_REGISTER = 1;
private Button btnLogin;
private EditText etAccount,etPassword;
private CheckBox cbRemember,cbAutoLogin;
private String userName = "dyy";
private String pass = "123";
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_login);
    getSupportActionBar().setTitle("登录");initView();initData();
    btnLogin.setOnClickListener(new View.OnClickListener() {
        public void onClick(View v) {String account =
etAccount.getText().toString();
String password = etPassword.getText().toString();
Log.d(TAG, "onClick-----" + account);
Log.d(TAG, "password-----" + password);
        if (TextUtils.isEmpty(userName)) {
Toast.makeText(LoginActivity.this, "还没有注册账号!",
Toast.LENGTH_LONG).show(); return; }
if(TextUtils.equals(account,userName)){if(TextUtils.equals(password,
pass)){Toast.makeText(LoginActivity.this,"登录成功!",
Toast.LENGTH_LONG).show();
SharedPreferences.Editor edit = spf.edit();
edit.putString("account", account);edit.putString("password", password);
edit.putBoolean("isRemember", true);
if(cbAutoLogin.isChecked()){edit.putBoolean("isLogin",true);}else{
edit.putBoolean("isLogin", false);}edit.apply();}else{SharedPreferences spf
= getSharedPreferences("spfRecord", MODE_PRIVATE);
SharedPreferences.Editor edit = spf.edit();
edit.putBoolean("isRemember", false);edit.apply();}
Intent intent = new Intent(LoginActivity.this, MainActivity.class);
intent.putExtra("account", account);startActivity(intent);
LoginActivity.this.finish(); }else {Toast.makeText(LoginActivity.this, "密码
错误!", Toast.LENGTH_LONG).show(); } } else
{ Toast.makeText(LoginActivity.this, "用户名错误!",
Toast.LENGTH_LONG).show();}}};
cbAutoLogin.setOnCheckedChangeListener(newCompoundButton.OnChe
ckedChangeListener() {public void
onCheckedChanged(CompoundButton buttonView, boolean isChecked)
{if (isChecked) { cbRemember.setChecked(true); } } });
cbRemember.setOnCheckedChangeListener(newCompoundButton.OnChe
ckedChangeListener

(){public void onCheckedChanged(CompoundButton buttonView, boolean
isChecked)
{if (!isChecked){ cbAutoLogin.setChecked(false); } } }); }
private void initView() { btnLogin =
findViewById(R.id.btn_login);
etAccount = findViewById(R.id.et_account);
etPassword = findViewById(R.id.et_password);
cbRemember = findViewById(R.id.cb_remember);
cbAutoLogin = findViewById(R.id.cb_auto_login); }
private void initData() {
SharedPreferences spf = getSharedPreferences("spfRecord",
MODE_PRIVATE);
boolean isRemember = spf.getBoolean("isRemember", false);
boolean isLogin = spf.getBoolean("isLogin", false);
String account = spf.getString("account", "");

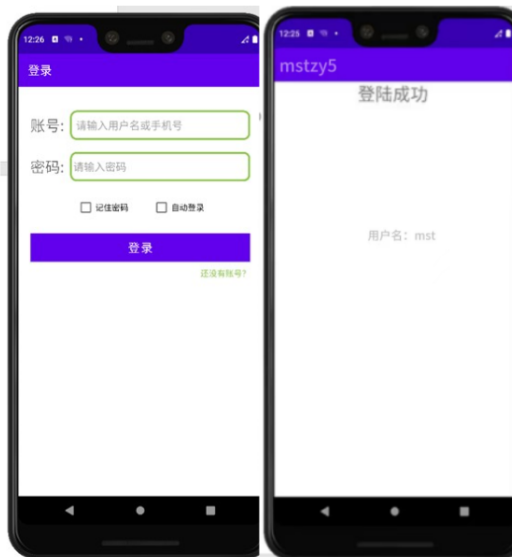
```

```

String password = spf.getString("password", "");
if (isLogin) {
    Intent intent = new Intent(LoginActivity.this,
MainActivity.class);
    intent.putExtra("account", account);
    startActivity(intent);
    LoginActivity.this.finish(); }
userName = account; pass = password;
if (isRemember) {etAccount.setText(account);
    etPassword.setText(password);
    cbRemember.setChecked(true);} }
public void toRegister(View view) {
    Intent intent = new Intent(this, RegisterActivity.class);
    startActivityForResult(intent, REQUEST_CODE_REGISTER); }
protected void onActivityResult(int requestCode, int resultCode,
@Nullable Intent data) { super.onActivityResult(requestCode, resultCode,
data);
    if (requestCode == REQUEST_CODE_REGISTER && resultCode
== RegisterActivity.RESULT_CODE_REGISTER && data != null) {
        Bundle extras = data.getExtras();
        String account = extras.getString("account", "");
        String password = extras.getString("password", "");
        etAccount.setText(account);
        etPassword.setText(password);
        userName = account;
        pass = password; } }}

```

运行结果：



第六章作业：

1.通过广播接收器实现两个 App 之间通过自定义广播进行数据通信。

部分关键代码：

```
//dyyzue61
//广播接收器
public void onReceive(Context context, Intent intent) {
    Toast.makeText(context,"
broadcast",Toast.LENGTH_LONG).show();}
//广播传递
public class MyBroadcastReceiver extends
BroadcastReceiver { public void
onReceive(Context context, Intent intent) {
    Toast.makeText(context,"broadcast",Toast.LENGTH_
LONG).show(); abortBroadcast();}}
//发送广播
Button button =
(Button)findViewById(R.id.button);
button.setOnClickListener(new
View.OnClickListener() { @Override
public void onClick(View view) {
    Intent intent = new
Intent("com.example.broadcast.MY_BROADCAST");
    sendOrderedBroadcast(intent,null);}
});
//dyyzue62 的广播接收器
public void onReceive(Context context, Intent intent) {
    Toast.makeText(context,"received in another
broadcast",Toast.LENGTH_LONG).show();}
}
//接收广播
public class CustomBroadReceiver extends
BroadcastReceiver{ private static final String
ACTION = "com.dyy.broadcast"; @Override
public void onReceive(Context context,
Intent intent) { if
(intent.getAction().equals(ACTION)){
    AlertDialog.Builder builder = new AlertDialog.Builder(context);
    .setMessage("收到上个的广播")
    AlertDialog dialog = (AlertDialog) builder.create();
    dialog.getWindow().setType(WindowManager.LayoutParams.TYPE_S
YSTEM_ALERT); dialog.show();}
}
}
```

运行结果：



第七章作业:

1. 使用 SQLite 实现记录登录时间的功能, 并且能够查询登录的历史记录。

部分关键代码:

```
//数据库
static final String CREATE_SQL[] = {"CREATE TABLE news (" +
    "_id INTEGER NOT NULL PRIMARY KEY\n" +
    "AUTOINCREMENT," +
    "title varchar," +
    + "content\n" +
    "varchar," +
    "keyword\n" +
    "varchar," +
    + "category\n" +
    "varchar," +
    + "author\n" +
    "INTEGER," +
    +
    "publish_time varchar" + ")",
    "CREATE TABLE user (" + "_id INTEGER NOT NULL PRIMARY KEY\n" +
    "AUTOINCREMENT," +
    "name varchar," +
    "password\n" +
    "varchar," + "email\n" +
    "varchar," +
    "phone varchar," +
    + "address\n" +
    "varchar" + ")",
    "INSERT INTO user VALUES (1,'dyy',123,123,'沈阳')",
    "INSERT INTO user VALUES (2,'dyy2',123,123,'北京')"};

public SQLiteDBHelper(@Nullable Context context, @Nullable String
name, int version) { super(context, name, null, version);}

@Override
public void onCreate(SQLiteDatabase db) {Log.i("sqlite____", "create
Database");for (int i = 0; i < CREATE_SQL.length; i++)
{db.execSQL(CREATE_SQL[i]);Log.i("sqlite____", "Finished Database");}

//登录
public class SQLiteLoginActivity extends
AppCompatActivity { SQLiteDBHelper
dbHelper;
EditText
etUsername;
EditText
etPassword;
```

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_sqlite_login); dbHelper = new
    SQLiteDBHelper(this,"sqlite.db",1);
    etUsername =
    findViewById(R.id.etUsername);
    etPassword =
    findViewById(R.id.etPassword);}
    public void onClick(View view) {
        String username = etUsername.getText().toString();
        String password = etPassword.getText().toString();Cursor cursor =
        db.query("user",new String[]{"name","password"}, "name=?",new
        String[]{username},null,null,null,"0,1");
        if(cursor.moveToNext()) {
            String dbPassword =
            cursor.getString(cursor.getColumnIndex("password"));
            cursor.close();if(password.equals(dbPassword)) {

                Intent intent = new Intent(this, SQLiteListActivity.class);
                startActivity(intent);return;}}cursor.close();
                Toast.makeText(this,"输入信息有误! ",Toast.LENGTH_LONG).show();}
@Override
public void
onDestroy() {
    super.onDestroy();
    dbHelper.close();}
//登陆成功跳转
ListView listView01;
SQLiteDBHelper
dbHelper;
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_sqlite_list);
    Toast.makeText(this,"登录成功
    ",Toast.LENGTH_LONG).show(); listView01 =
    findViewById(R.id.listView01);
    dbHelper = new SQLiteDBHelper(this,"sqlite.db",1);
    SQLiteDatabase db = dbHelper.getReadableDatabase();
    Cursor cursor = db.query("user",new
    String[]{"_id","name"},null,null,null,null,null);
    SimpleCursorAdapter adapter = new SimpleCursorAdapter(this,

```

```
android.R.layout.simple_list_item_1,cursor,new  
String[]{"_id","name"}, new  
int[]{android.R.id.text1,android.R.id.text1},);  
listView01.setAdapter(adapter); }  
运行结果：
```



第八章作业：

- 1、使用 Camera2 实现定时拍照的功能。部分关键代码：

```
private static final int
CAMERA_ID = 0; private
static final int CAMERA_ID
= 1;
private static final String TAG =
MainActivity.class.getSimpleName(); protected void
onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main); }
protected void onPause() {
super.onPause(); stopPreview(); }
private void startPreview(){if (mCamera !=
null || !mIsSurfaceCreated) { Log.d(TAG, "sta rtPreview will return");
return; }
mCamera = Camera.open(CAMERA_ID);
Camera.Parameters parameters =
mCamera.getParameters(); int width =
getResources().getDisplayMetrics().widthPixel
s; int height =
getResources().getDisplayMetrics().heightPixe
ls;
Camera.Size size = getBestPreviewSize(width,
height, parameters); if (size != null)
{parameters.setPreviewSize(size.width,
size.height);
parameters.setPictureSize(size.width,
size.height); }
parameters.setFocusMode(Camera.Parameters.FOC
US_MODE_AUTO);
parameters.setPreviewFrameRate(20);
mCamera.setDisplayOrientation(90);
mCamera.setParameters(parameters);
```

```

        try {mCamera.setPreviewDisplay(mHolder); } catch
        (Exception e) { Log.d(TAG, e.getMessage()); }
        mCamera.startPreview(); } } }

private Camera.Size getBestPreviewSize(int width,
    int height, Camera.Parameters parameters) {
    Camera.Size result = null;
    for (Camera.Size size : parameters.getSupportedPreviewSizes()) {
        if (size.width <= width && size.height <= height) {
            if (result == null) { result =
            size; } else { int resultArea
            = result.width * result.height;
            int newArea = size.width *
            size.height;
            if (newArea > resultArea) { result = size; } } } } return result; }

    public void onClick(View v) { switch (v.getId()) { case R.id.start:
    if (!mIsTimerRunning){ mIsTimerRunning = true;
    mHandler.post(timerRunnable);} break;} private Runnable
    timerRunnable = new Runnable() {
    public void run() { if (mCurrentTimer > 0) {
    mTvCountDown.setText(mCurrentTimer + "");
    mCurrentTimer--; mHandler.postDelayed(timerRunnable,
    1000); } else { mTvCountDown.setText("");
    mCamera.takePicture(null, null, null, MainActivity.this);
    playSound(); mIsTimerRunning = false; mCurrentTimer =
    10; } } };

    public void onPictureTaken(byte[] data, Camera camera) {

try { FileOutputStream fos = new FileOutputStream(new File
                                                                    (Environment.getExternalStorageDirectory() + File.separator + System.currentTimeMillis()
                                                                    + ".png"));
    }

```

运行结果：

